

Aggression Analysis in YouTube News Comments

Team:- Neuronauts

Abstract—

1. VOLATILITY-DRIVEN REGIME DYNAMICS IN LIVE YOUTUBE COMMENT STREAMS

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1.1. Introduction

Live digital comment streams provide a real-time setting for observing collective emotional behavior. A central question in online social dynamics is whether anger spreads through self-reinforcing cascades or behaves as a volatile but self-correcting signal.

Traditional analyses focus on average aggression levels. However, mean intensity does not capture instability or rapid shifts in sentiment. From a cognitive standpoint, volatility may better reflect disagreement, uncertainty, and emotional reactivity.

This study asks:

Does emotional volatility predict short-term regime shifts in collective anger, and what does this imply about cognitive processes in digital crowds?

1.2. Data and Aggregation

The dataset was collected from a **live YouTube news** broadcast covering developments related to the **Russia-Ukraine conflict (TXdLuN-JedcM)**. The broadcast involved real-time reporting and unfolding geopolitical updates, naturally generating mixed emotional responses among viewers.

Unlike static uploaded videos, **live broadcasts create a dynamic interaction environment** in which viewers respond not only to the video content but also to preceding comments. Emotional expression in this setting is therefore sequentially dependent and temporally structured. This makes live chat particularly suitable for studying **continuous emotional regime shifts** rather than isolated sentiment snapshots.

1.2.1. Preprocessing

Transformer-Based Emotion Extraction

Each comment was processed using the pre-trained transformer model **j-hartmann/emotion-english-distilroberta-base**. This model is a DistilRoBERTa-based architecture fine-tuned for multi-class emotion classification in English text. It produces probabilistic outputs across discrete emotional categories.

From these outputs, the **anger probability score** was extracted and treated as a continuous measure of aggression intensity.

The use of a transformer-based model is methodologically appropriate for three reasons. First, transformer architectures capture contextual semantics rather than relying on keyword-based matching. Second, they are robust to informal language patterns commonly found in live chat environments. Third, fine-tuning on emotion-labeled corpora allows the model to detect subtle affective cues, including implicit hostility or context-dependent emotional framing.

This approach enables high-resolution emotional tracking **suitable for modeling continuous regime shifts** in collective sentiment.

Illustrative Examples. For example, the comments **“russia didn’t respect anything”, “red army kicked by trans army, u.s.a. and Europe are hypocriser** received a **high anger probability score**, reflecting strongly accusatory language. In contrast, comments such as **“RIP JEWKRAINE”, “hahaha have fun hahaha”, and “wow nice”** were assigned comparatively **low anger probabilities**. These contrasting outputs illustrate the model’s ability to differentiate between

overt hostility and comments expressing sarcasm, humor, or neutral tone. Such probabilistic variation forms the basis for constructing a continuous anger signal prior to temporal aggregation.^z

Temporal Block Aggregation

To preserve temporal dynamics while reducing individual-level noise, comments were aggregated into **non-overlapping 5-minute blocks**.

Let A_t denote the mean anger score in block t .

This aggregation performs two functions:

- It smooths random fluctuations at the individual-comment level.
- It preserves short-term volatility patterns necessary for regime modeling.

The choice of 5-minute granularity balances **responsiveness and stability**. **Shorter windows** would introduce excessive noise, while longer windows would obscure rapid emotional transitions.

The live, high-salience, and interaction-driven nature of the broadcast makes it an appropriate empirical setting for modeling **continuous volatility dynamics** and testing whether collective anger behaves as contagion or bounded reactivity.

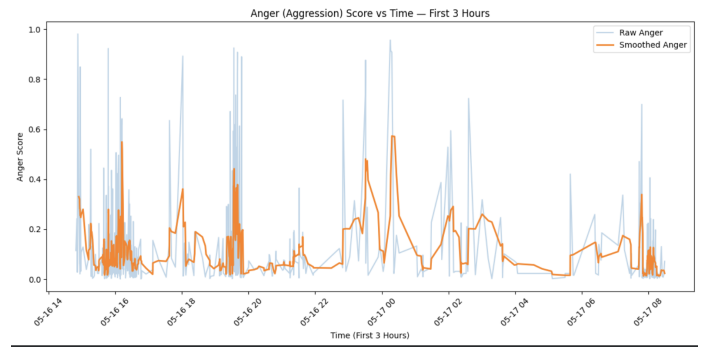


Figure 1. Raw and smoothed anger dynamics.

1.3. Volatility-Based Regime Model

This subsection introduces the volatility-driven regime framework used to model short-term transitions in collective anger. Rather than predicting raw anger intensity, the model estimates transitions between emotional regimes.

1.3.1. State Definition

Let A_t denote the mean anger score in 5-minute block t . A binary emotional regime variable is defined as:

$$S_t = \begin{cases} 1 & \text{if } A_t > \text{median}(A) \\ 0 & \text{otherwise} \end{cases}$$

This separates the time series into **high-anger** and **low-anger** regimes. The objective is to model transitions between these regimes.

1.3.2. Volatility Measure

Emotional volatility is defined as the rolling standard deviation of anger over the previous six blocks (30 minutes):

$$Vol_t = \text{Std}(A_{t-5}, \dots, A_t)$$

Volatility captures short-term instability and disagreement within the comment stream. Unlike mean intensity, volatility reflects fluctuations and uncertainty in emotional expression.

1.3.3. Feature Construction

To model regime transitions, the following predictors were constructed:

- S_t — Current emotional regime (state persistence)
- A_t — Current anger intensity
- ΔA_t — Momentum (block-to-block change)
- Vol_t — Emotional volatility
- $Z_t = \frac{A_t - \mu_t}{\sigma_t}$ — Distance from local baseline

These features jointly represent persistence, intensity, instability, and deviation dynamics.

1.3.4. Transition Model

The probability of entering a high-anger regime in the next block is modeled using logistic regression:

$$P(S_{t+1} = 1 \mid S_t, A_t, \Delta A_t, Vol_t, Z_t)$$

This formulation treats emotional dynamics as a state-transition process rather than a purely continuous regression problem.

1.3.5. Results

Single-Split Test Performance

Using a chronological 70/30 train-test split, the model achieved:

- **Accuracy:** 0.658
- **Log-Likelihood:** -0.650
- **AIC:** 13.30
- **BIC:** 23.13

An accuracy of 0.658 exceeds random chance (0.50), indicating that short-term emotional regime transitions are partially predictable. However, the log-likelihood suggests that transitions remain probabilistic rather than deterministic.

Model Coefficients and Interpretation

The logistic regression model estimating

$$P(S_{t+1} = 1 \mid S_t, A_t, \Delta A_t, Vol_t, Z_t)$$

produced the following coefficients:

- S_t : 0.747
- A_t : -0.445
- ΔA_t : -1.079
- Vol_t : 0.175
- Z_t : 0.276
- Intercept: -0.255

State Persistence ($S_t = 0.747$). The positive coefficient on the current state indicates **emotional persistence**. When the system is already in a high-anger regime, the likelihood of remaining in that regime increases in the subsequent interval. This reflects short-term inertia in collective emotional dynamics.

Current Anger Level ($A_t = -0.445$). The negative coefficient on current anger suggests a **mean-reversion tendency**. Extremely elevated anger levels are slightly less likely to be followed by continued escalation. This indicates that peaks in emotional intensity often precede moderation.

Momentum ($\Delta A_t = -1.079$). The strongly negative coefficient on momentum is particularly informative. Rapid increases in anger are associated with a lower probability of remaining in a high-anger regime in the next block. This pattern suggests **emotional overshooting**: sharp escalations tend to trigger corrective stabilization rather than runaway amplification.

Volatility ($Vol_t = 0.175$). The positive coefficient on volatility indicates that **emotional instability increases the likelihood of regime transition**. Periods of disagreement and fluctuation make the collective state more prone to change. This supports the central hypothesis that volatility is structurally informative in modeling emotional dynamics.

Distance from Baseline ($Z_t = 0.276$). The positive coefficient on deviation from the local mean suggests that large departures from recent norms destabilize the current regime. However, such deviations do not necessarily imply sustained cascades; instead, they increase the probability of short-term regime shifts.

Overall Structural Pattern. Taken together, the coefficients reveal a coherent structure: emotional states exhibit persistence, but sharp increases tend to correct rather than amplify. Volatility and deviation promote instability, yet momentum signals correction. This pattern supports a bounded-volatility interpretation rather than strong contagion-based cascade dynamics.

Cross-Validation Performance

To assess robustness, time-series cross-validation with four sequential folds was conducted:

- Fold 1: Accuracy = 0.5385, Log-Likelihood = -0.6658
- Fold 2: Accuracy = 0.5769, Log-Likelihood = -0.8984
- Fold 3: Accuracy = 0.5769, Log-Likelihood = -0.6950
- Fold 4: Accuracy = 0.6923, Log-Likelihood = -0.6615

Overall metrics:

- Mean Accuracy: 0.5962
- Standard Deviation (Accuracy): 0.0577
- Mean Log-Likelihood: -0.7302

Cross-validation confirms moderate but stable predictive performance. While predictive strength varies across folds, accuracy consistently exceeds random chance, indicating structured -though probabilistic- emotional regime dynamics.

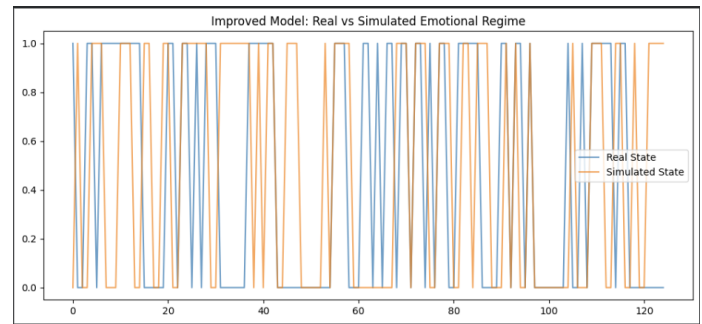


Figure 2. Real vs Simulated Emotional Regime

1.4. Cascade Threshold Test

Model Overview.

To examine whether collective anger follows contagion-style escalation, a threshold-based cascade model was implemented. The model is inspired by **Granovetter's threshold theory**, which proposes that collective cascades occur when individual activation thresholds are sequentially crossed, producing self-reinforcing group behavior.

In this study, the threshold concept is adapted to emotional dynamics. Instead of modeling participation in collective action, we model **emotional activation**. A cascade event is defined as a statistically significant upward deviation in collective anger relative to its recent baseline.

Emotional activation is operationalized using a rolling Z-score:

$$Z_t = \frac{A_t - \mu_t}{\sigma_t}$$

where A_t is the 5-minute mean anger level, and μ_t and σ_t are the rolling mean and standard deviation over the previous 30 minutes.

A **threshold crossing** occurs when Z_t moves above a predefined standard deviation cutoff. A **sustained cascade** is recorded only if anger remains elevated above its rolling mean for the next two blocks. This definition distinguishes between temporary spikes and persistent contagion.

1.5 Standard Deviation Threshold

Using a high activation threshold of $Z_t > 1.5$:

- Total upward threshold crossings: 11
- Sustained elevations: 1
- Cascade probability: $\approx 9\%$

Although extreme emotional spikes occurred, only a single event remained elevated across subsequent intervals. This indicates **weak persistence of high-intensity emotional activation**.

1 Standard Deviation Threshold

Using a lower activation threshold of $Z_t > 1.0$:

- Total upward threshold crossings: 17
- Sustained elevations: 1
- Cascade probability: $\approx 5.9\%$

Lowering the threshold increased the number of activation events, suggesting that online participants frequently respond to rising emotional signals. However, the extremely low persistence rate indicates that such responses rarely evolve into sustained cascades.

Comparative Interpretation

Across both threshold levels, **activation events were common but sustained cascades were rare**. Even when the threshold was relaxed to 1 SD, the probability of prolonged escalation remained below 6%.

These results suggest that online users may temporarily follow emotional spikes, but collective anger does not typically develop into self-reinforcing amplification. Emotional dynamics in live comment streams therefore resemble **reactive synchronization** rather than **contagion-driven escalation**.

Overall, the evidence supports a **bounded-instability interpretation** of collective anger rather than a strong cascade model.

1.5. Mean-Reversion Analysis

Conceptual Motivation.

In quantitative finance, **Bollinger Bands** are commonly used to test whether a price series exhibits **mean-reverting** or **trend-following** behavior. The method constructs dynamic upper and lower bands around a moving average using rolling standard deviations. When prices deviate significantly from the local mean, traders test whether the series tends to revert back toward baseline or continue trending.

Although originally developed for financial markets, the framework can be applied to any continuous time series. In this study, collective anger is treated as a **dynamic signal** analogous to a price series. Since anger is measured as a continuous probability score aggregated over time, it is appropriate to test whether emotional spikes behave as mean-reverting deviations or as persistent escalations.

Methodology

Let A_t denote the 5-minute mean anger level. A rolling baseline μ_t and rolling standard deviation σ_t were computed over the previous six blocks (30 minutes). Upper and lower bands were defined as:

$$\text{Upper Band} = \mu_t + 1.5\sigma_t$$

$$\text{Lower Band} = \mu_t - 1.5\sigma_t$$

A **spike event** was recorded when A_t crossed above the upper band. A successful mean-reversion event was defined as a return of anger toward the rolling mean within subsequent intervals.

This operationalization allows testing whether emotional surges tend to correct or amplify.

Results

Across 20 detected spike events:

- **Reversion Success Rate: 85%**
- Majority of spikes returned toward baseline within short horizons

Only a small fraction of spikes developed into prolonged elevation.

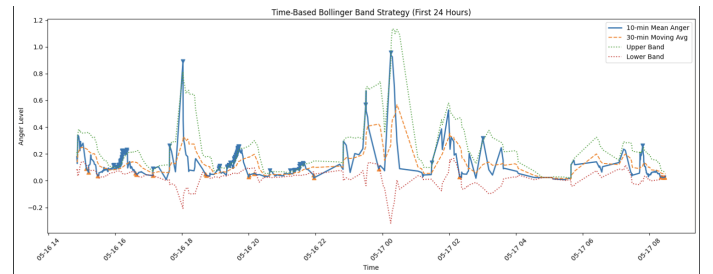


Figure 3. Mean-reversion behavior of collective anger using Bollinger Bands. The figure illustrates spike events where anger crosses the upper band and subsequently returns toward the rolling baseline.

Interpretation

The high reversion rate strongly suggests that collective anger behaves as a **mean-reverting process** rather than a trend-following cascade.

Spikes appear to represent **temporary emotional overshooting** rather than sustained amplification. Collective emotional intensity rises rapidly but subsequently corrects toward equilibrium.

Cognitive Implications

From a cognitive perspective, this pattern aligns with models of **bounded affective reactivity**. Initial comments may trigger rapid emotional alignment driven by fast, intuitive processes. However, corrective mechanisms — including counter-arguments, informational updates, attention shifts, and fatigue — appear to stabilize the system.

The findings indicate that digital crowds exhibit **short-term synchronization without long-term emotional locking**. Anger functions as a volatile signal that fluctuates around a baseline rather than evolving into persistent collective escalation.

Together with the cascade threshold analysis, the mean-reversion results reinforce the interpretation that collective anger in live comment streams is characterized by **reactive volatility with stabilization**, rather than contagion-driven runaway dynamics.

1.6. Conclusion

This study investigated whether collective anger in live YouTube comment streams follows self-reinforcing cascade dynamics or exhibits bounded volatility. Across regime modeling, threshold-based cascade testing, and mean-reversion analysis, the evidence consistently favors the latter.

Emotional spikes occur frequently and users often react to rising anger signals. However, escalation rarely persists. Both low cascade probabilities and high mean-reversion rates indicate that collective anger typically stabilizes rather than amplifies.

These findings suggest that digital crowds exhibit **rapid emotional synchronization without sustained contagion**. Collective anger behaves as a **reactive but self-correcting process** - volatile, yet structurally constrained.

Overall, live comment streams display **rapid amplification followed by stabilization**, indicating that collective anger online is **volatile but cognitively regulated rather than contagion-driven**.

2. LOCATION-BASED ANALYSIS

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3. ABSTRACT

This study develops a dynamic probabilistic framework to examine how latent cognitive aggression evolves over time. The model evaluates the influence of exposure, prior mental states, regional differences (Venezuela, Global, and USA), gender differences. Model validity is assessed using likelihood-based measures and classification performance metrics.

Building on Assignment-1, which defined antecedents, latent states, behaviors, and consequences, this study models the temporal evolution of hidden aggression states using a logistic Markov framework. Parameters are estimated via Maximum Likelihood Estimation (MLE), and performance is evaluated using goodness-of-fit statistics and predictive metrics.

- **Antecedent (X_t):** Exposure to aggressive tone
- **Latent State (S_t):** Binary aggression state (0 = non-aggressive, 1 = aggressive)
- **Behavior (B_t):** Expressed aggressive comment
- **Consequence (C_t):** Engagement amplification

The temporal transition is modeled as:

$$P(S_{t+1} \mid S_t, X_t)$$

4. METHODOLOGY

4.1. Model Specification

The transition probability follows a logistic formulation:

$$P(S_{t+1} = 1 \mid S_t, X_t) = \sigma(\alpha + \beta X_t + \gamma S_t)$$

where

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- α : Baseline activation (intercept)
- β : Sensitivity to external exposure
- γ : State persistence parameter

4.2. Rolling Exposure Variable

To capture cumulative exposure effects, the stimulus variable X_t is constructed using a rolling window of 50 comments. The rolling mean of past behavioral indicators generates a continuous exposure measure, allowing gradual build-up of environmental influence.

4.3. Parameter Estimation and Evaluation

Model parameters are estimated using Maximum Likelihood Estimation with the L-BFGS-B optimization algorithm. Model performance is assessed using:

- **McFadden's Pseudo R^2**
- **AIC and BIC**
- **Classification Metrics:** Accuracy, Precision, Recall, F1-score, ROC-AUC

The log-likelihood function is defined as:

$$\mathcal{L}(\theta) = - \sum \log P(S_{t+1} \mid S_t, X_t)$$

5. RESULTS

5.1. Venezuela

Estimated Parameters

Parameter	Value
α	-3.9107
β	11.7074
γ	-0.8042
AIC	1521.53
BIC	1541.64

Model Performance

Metric	Value
Accuracy	0.6300
Precision	0.0404
Recall	0.5417
F1 Score	0.0752
ROC-AUC	0.6069

Goodness of Fit

- Log-Likelihood (Model): -757.76
- Log-Likelihood (Null): -767.62
- McFadden Pseudo R^2 : 0.0128

The pseudo- R^2 indicates modest improvement over the intercept-only model, suggesting limited but meaningful explanatory power.

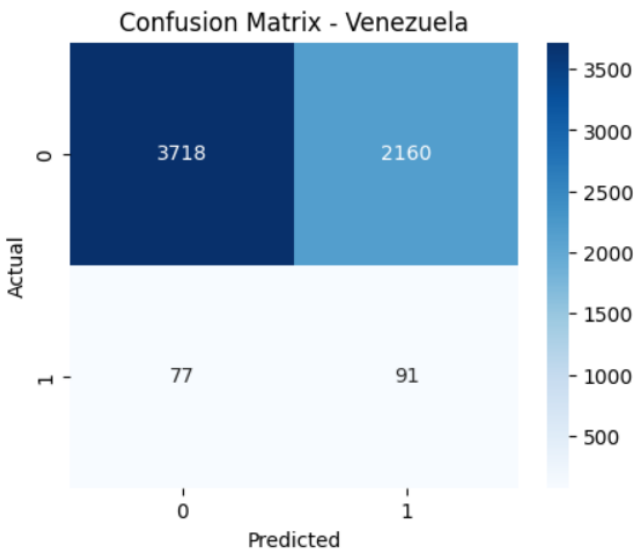


Figure 4. Confusion Matrix – Venezuela

From/To	Calm	Aggressive
Calm	0.971929227628445	0.02807077237155495
Aggressive	0.9821428571428571	0.017857142857142856

Figure 5. Transition Matrix – Venezuela

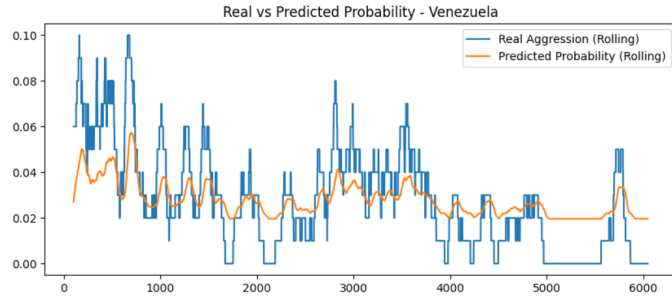


Figure 6. Real vs Predicted Probabilities – Venezuela

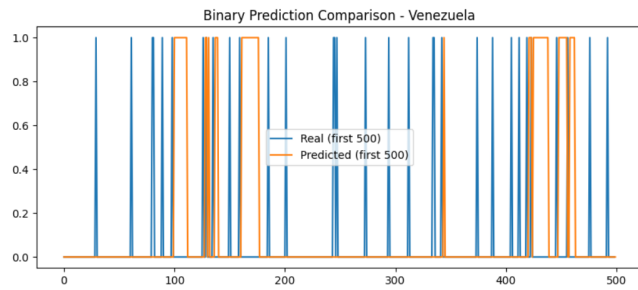


Figure 7. Binary Prediction Comparison – Venezuela

5.2. Global

Estimated Parameters

Parameter	Value
α	-3.2149
β	6.2007
γ	0.3056
AIC	1875.71
BIC	1894.82

Model Performance

Metric	Value
Accuracy	0.5386
Precision	0.0770
Recall	0.6437
F1 Score	0.1376
ROC-AUC	0.5955

Goodness of Fit

- Log-Likelihood (Model): -934.85
- Log-Likelihood (Null): -946.56
- McFadden Pseudo R^2 : 0.0124

The model shows slight improvement over the null model, with persistence playing a more visible role compared to Venezuela.

Visual Diagnostics

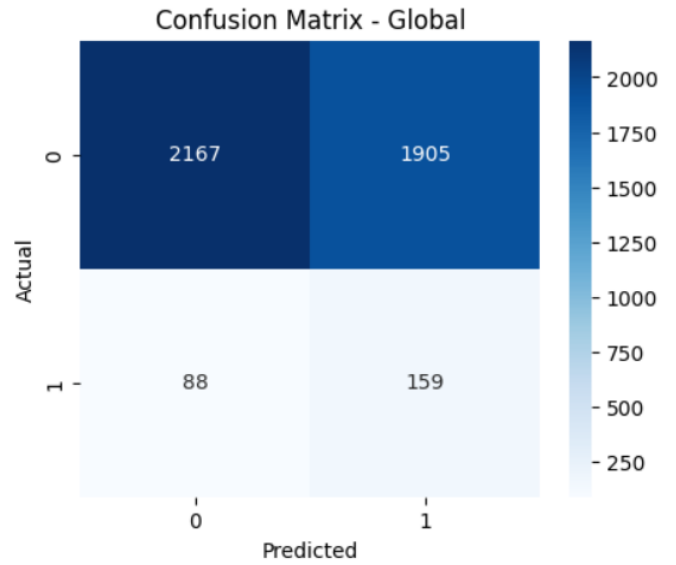


Figure 8. Confusion Matrix – Global

From/To	Calm	Aggressive
Calm	0.9447445972495089	0.05525540275049116
Aggressive	0.9109311740890689	0.08906882591093117

Figure 9. Transition Matrix – Global

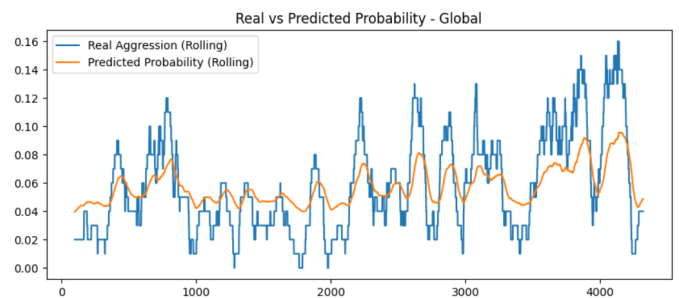


Figure 10. Real vs Predicted Probabilities – Global

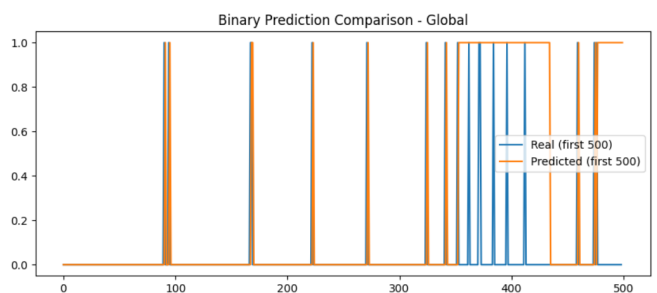


Figure 11. Binary Prediction Comparison – Global

5.3. USA

Estimated Parameters

Parameter	Value
α	-2.7872
β	-8.4375
γ	-1.1308
AIC	684.15
BIC	700.94

Model Performance

Metric	Value
Accuracy	0.3976
Precision	0.0492
Recall	0.7439
F1 Score	0.0923
ROC-AUC	0.5645

Goodness of Fit

- Log-Likelihood (Model): -339.07
- Log-Likelihood (Null): -341.88
- McFadden Pseudo R^2 : 0.0082

The negative exposure coefficient indicates inhibitory dynamics, while overall model fit remains modest.

Visual Diagnostics

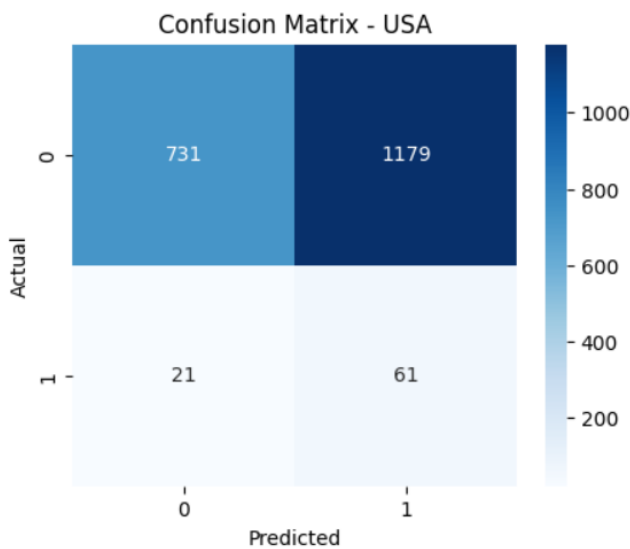


Figure 12. Confusion Matrix – USA

From/To ▼	Calm	Aggressive
Calm	0.9575916230366492	0.042408376963350786
Aggressive	0.9878048780487805	0.012195121951219513

Figure 13. Transition Matrix – USA

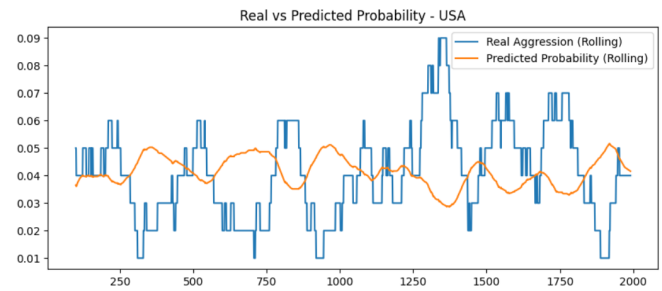


Figure 14. Real vs Predicted Probabilities – USA

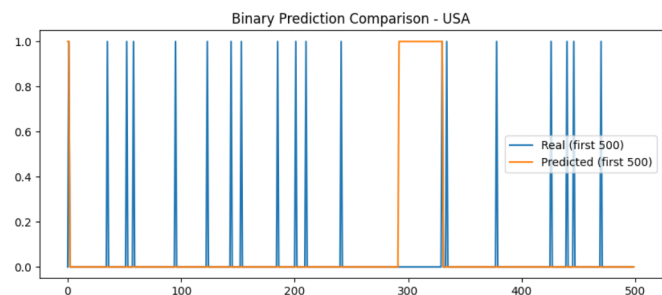
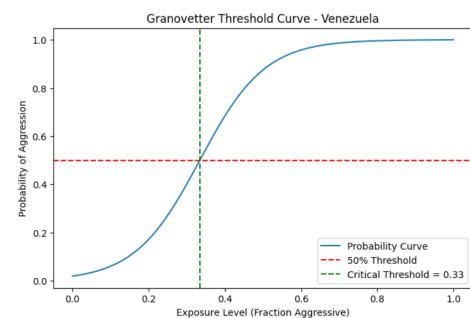


Figure 15. Binary Prediction Comparison – USA

6. GRANOVETTER CURVES

The Granovetter Curves below illustrate the "tipping point" for each region—the level of environmental exposure (X) required to make a state transition likely.

6.1. Venezuela



Critical Threshold for Venezuela: 0.334

Figure 16. Granovetter Threshold Curve – Venezuela (Critical Threshold: 0.334)

6.2. Global

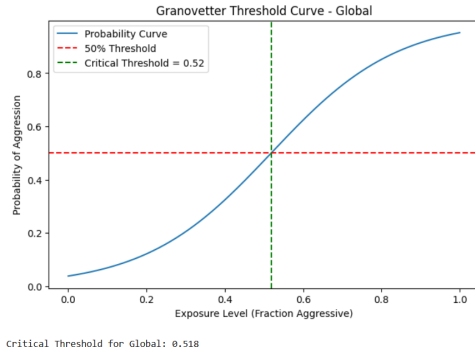


Figure 17. Granovetter Threshold Curve – Global (Critical Threshold: 0.518)

6.3. USA

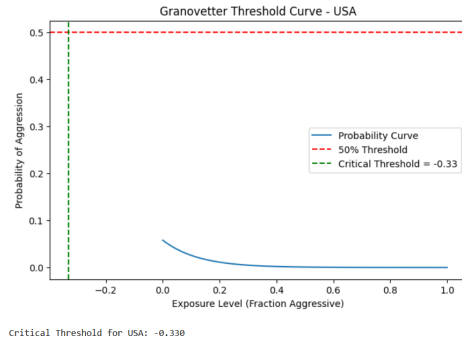


Figure 18. Granovetter Threshold Curve – USA (Inhibitory/Negative Response Curve)

7. REGIONAL COMPARATIVE ANALYSIS

This section combines the mathematical model parameters (α, β, γ) with behavioral interpretations to explain how aggression evolves differently across locations.

7.1. Venezuela

- **Mathematical Profile:** Lowest baseline ($\alpha = -3.911$) but highest sensitivity ($\beta = 11.707$) and negative persistence ($\gamma = -0.804$).
- **Analysis:** Venezuela represents a highly sensitive environment. While inherent aggression is low, users are extremely susceptible to the “vibe” of the conversation. They are easily “triggered” by external exposure, leading to a sharp, immediate response.
- **Tipping Point:** The **Critical Threshold is 33.4%**. It only takes a small amount of environmental aggression to flip a user’s state. However, the negative γ indicates that these outbursts are short-lived; users “cool off” as quickly as they flared up.

7.2. Global

- **Mathematical Profile:** Moderate sensitivity ($\beta = 6.201$) and unique positive persistence ($\gamma = 0.306$).
- **Analysis:** Unlike specific countries, global data reveals an “echo chamber” effect. It is the only case where aggression is “sticky.” Once a user enters an aggressive state, the mood carries over and reinforces itself in the next interaction.

- **Tipping Point:** The **Critical Threshold is 51.8%**. The community is more resilient than Venezuela, requiring a majority of aggressive exposure to shift the collective state, but once that shift happens, the state is much harder to break.

7.3. USA

- **Mathematical Profile:** Highest baseline ($\alpha = -2.787$) and negative sensitivity ($\beta = -8.437$).
- **Analysis:** The USA shows a high level of “simmering” internal tension even without external triggers. However, the response to stimulus is *inhibitory*. As the environment becomes more toxic, users actually become *less* likely to transition into aggression.
- **Interpretation:** This suggests “outrage fatigue” or active avoidance. When exposure becomes too high, users likely disengage or “tune out” rather than reacting, possibly due to more effective platform moderation or user exhaustion.

8. GENDER-BASED ANALYSIS

Building on Assignment–1, which defined antecedents, latent states, behaviors, and consequences, this study models the temporal evolution of hidden aggression states using a logistic Markov framework. Parameters are estimated via Maximum Likelihood Estimation (MLE), and performance is evaluated using goodness-of-fit statistics and predictive metrics.

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The temporal transition is modeled as:

$$P(S_{t+1} = 1 | S_t, X_t)$$

9. METHODOLOGY

9.1. Model Specification

The transition probability follows a logistic formulation:

$$P(S_{t+1} = 1 | S_t, X_t) = \sigma(\alpha + \beta X_t + \gamma S_t)$$

where

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- α : Baseline activation (intercept)
- β : Sensitivity to external exposure
- γ : State persistence parameter

9.2. Rolling Exposure Variable

To capture cumulative exposure effects, the stimulus variable X_t is constructed using a rolling window of 50 comments. The rolling mean of past behavioral indicators generates a continuous exposure measure, allowing gradual build-up of environmental influence.

9.3. Parameter Estimation and Evaluation

Model parameters are estimated using Maximum Likelihood Estimation with the L-BFGS-B optimization algorithm. Model performance is assessed using:

- **McFadden’s Pseudo R^2**
- **AIC and BIC**

- **Classification Metrics:** Accuracy, Precision, Recall, F1-score, ROC-AUC

The log-likelihood function is defined as:

$$\mathcal{L}(\theta) = -\sum \log P(S_{t+1} | S_t, X_t)$$

10. RESULTS

10.1. Males

Estimated Parameters

Parameter	Value
α	-3.366
β	6.649
γ	0.015
AIC	2949.023
BIC	2969.964

Model Performance

Metric	Value
Accuracy	0.4319
Precision	0.0547
Recall	0.696
F1 Score	0.1015
ROC-AUC	0.5744

Goodness of Fit

- Log-Likelihood (Model): -1471.511
- Log-Likelihood (Null): -1483.862
- McFadden Pseudo R^2 : 0.008

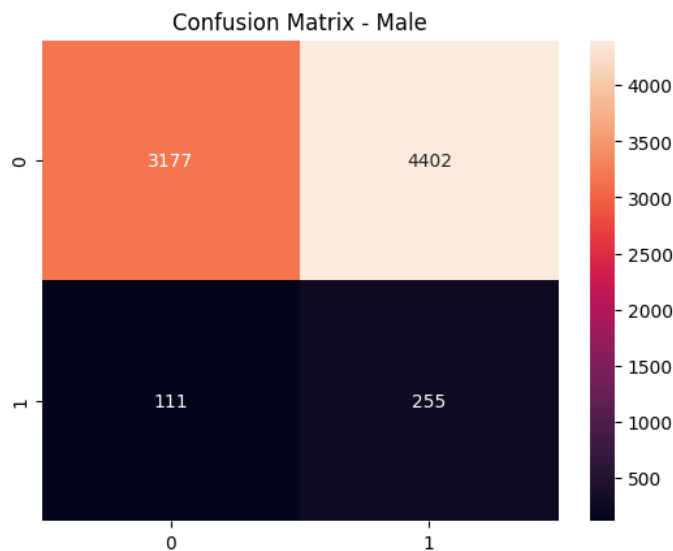


Figure 19. Confusion Matrix – Male

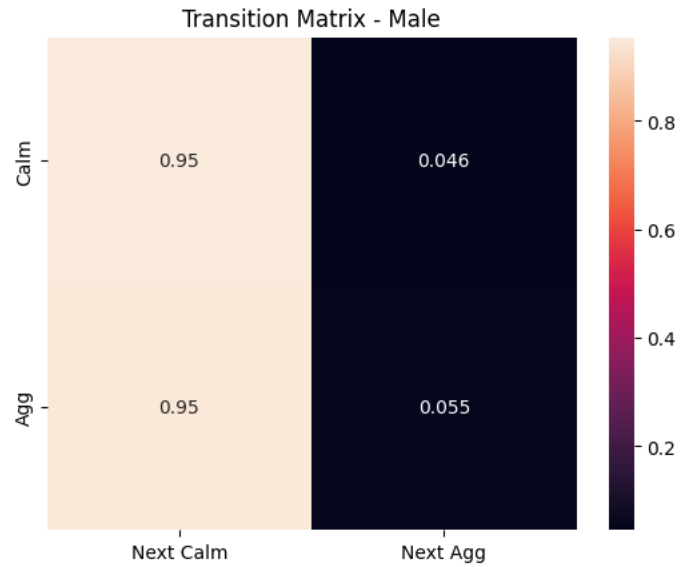


Figure 20. Transition Matrix – Male

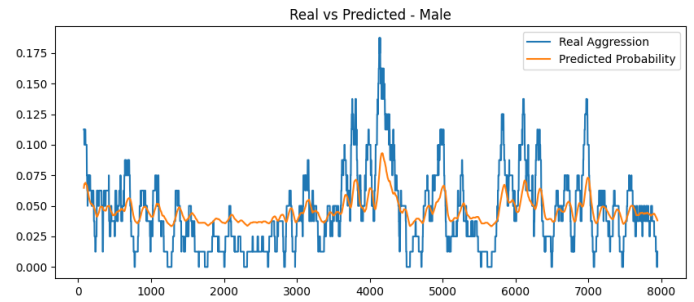


Figure 21. Real Vs Predicted Probabilities – Male

10.2. Females

Estimated Parameters

Parameter	Value
α	-3.082
β	2.047
γ	-0.259
AIC	253.186
BIC	266.542

Model Performance

Metric	Value
Accuracy	0.5565
Precision	0.0565
Recall	0.5161
F1 Score	0.1019
ROC-AUC	0.5239

Goodness of Fit

- Log-Likelihood (Model): -123.593
- Log-Likelihood (Null): -123.739
- McFadden Pseudo R^2 : 0.0011

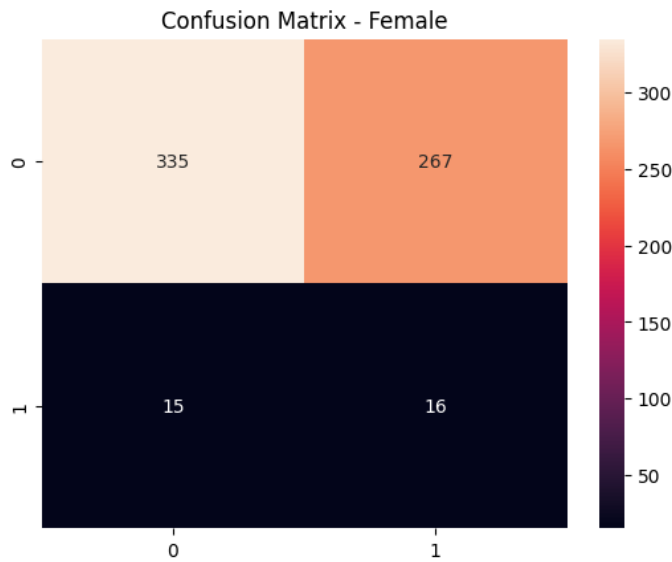


Figure 22. Confusion Matrix – Female

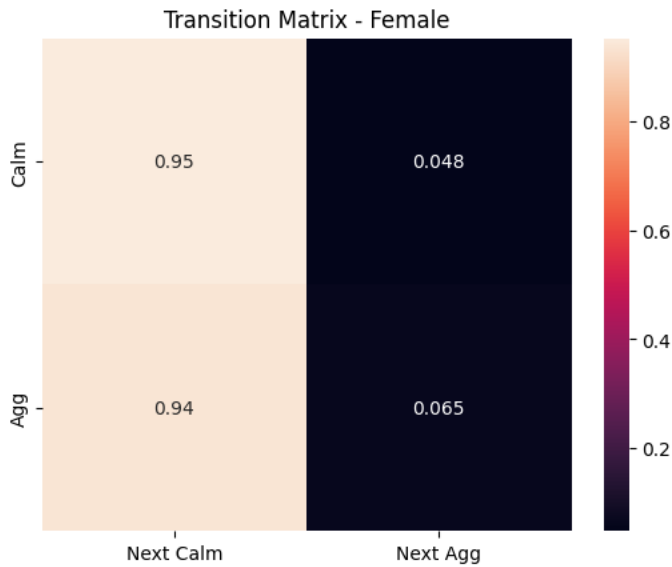


Figure 23. Transition Matrix – Female

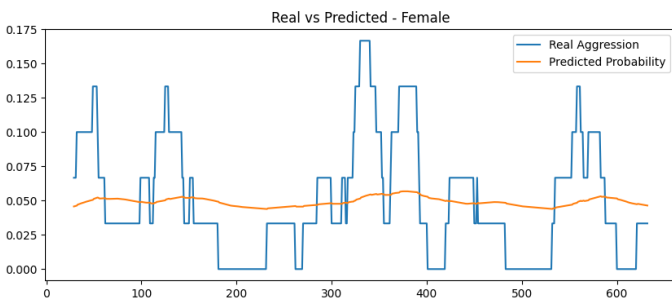


Figure 24. Real Vs Predicted Probabilities – Female

11. GRANOVETTER CURVES

The Granovetter Curves below illustrate the "tipping point" for each gender i.e., male and female, the level of environmental exposure (X) required to make a state transition likely.

11.1. Male

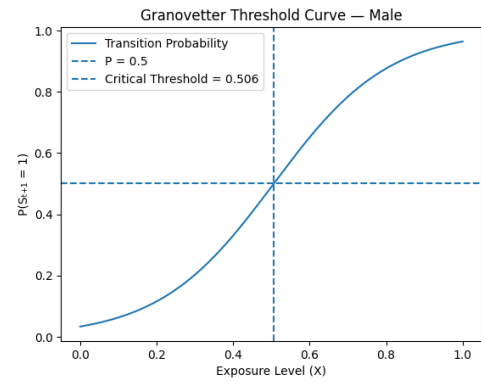


Figure 25. Granovetter Threshold Curve – Male (Critical Threshold: 0.5062)

11.2. Female

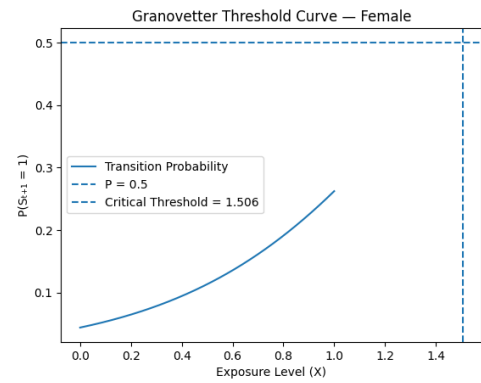


Figure 26. Granovetter Threshold Curve – Female (1.5056)

11.3. Simulation of Behavioral Trajectories

To examine the long-term evolution of latent aggression states, behavioral trajectories were simulated using the estimated Markovian transition parameters. The simulation iteratively updates the latent state based on the probabilistic transition function:

$$P(S_{t+1} = 1 \mid S_t, X_t) = \sigma(\alpha + \beta X_t - \delta S_t)$$

where the next state is sampled from the predicted transition probability. These simulations illustrate the stability, volatility, and persistence characteristics of aggression dynamics across genders.

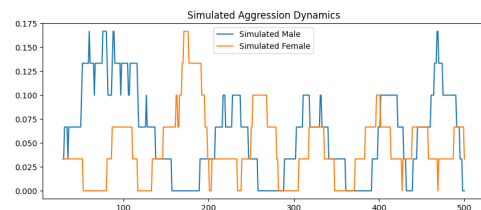


Figure 27. Simulation Plot

11.4. Analysis and Interpretation

11.4.1. Transition Dynamics

The transition matrices for male and female users reveal highly stable behavioral dynamics. For both genders, the probability of remaining

in the non-aggressive (calm) state is approximately 0.95, indicating strong persistence of non-aggressive behavior.

For aggressive states, the transition probabilities show that aggression is relatively short-lived. Female users exhibit a slightly lower probability of remaining aggressive (0.065) compared to males (0.055), suggesting marginally faster recovery from aggression. This implies that while aggression occurs, users tend to revert quickly to calmer states.

Overall, the dominance of calm-to-calm transitions highlights that aggressive comments represent intermittent deviations rather than sustained behavioral regimes.

11.4.2. Granovetter Threshold Analysis

Granovetter threshold curves illustrate the exposure level required to trigger an aggression state transition. The male model exhibits a critical threshold near 0.506, indicating that aggression becomes likely when exposure crosses roughly 50% aggressive content.

In contrast, the female model produces a much higher threshold (1.506), which lies outside the feasible exposure range. This reflects that predicted transition probabilities for females remain below 0.5 even at maximum exposure. Psychologically, this suggests greater resistance to aggression escalation among female users, with aggressive transitions occurring less readily in response to environmental stimuli.

The disparity between thresholds implies stronger stimulus sensitivity among males relative to females.

11.4.3. Real vs Predicted Probabilities

Comparison of real and predicted aggression probabilities shows distinct temporal characteristics. Male aggression trajectories display noticeable fluctuations, while predicted probabilities capture the overall trend but remain smoother, reflecting the probabilistic nature of the model.

Female aggression dynamics exhibit lower amplitude variations. The predicted probability curve remains relatively stable, indicating conservative estimation consistent with lower aggression base rates and reduced volatility.

The smoother predicted curves for both genders indicate that the model estimates expected behavioral tendencies rather than reproducing instantaneous noise.

11.4.4. Classification Performance

Confusion matrices reveal that the model correctly classifies the majority of calm states but struggles with aggressive state detection. This is consistent with the imbalanced distribution of aggression events.

Despite modest predictive metrics (ROC-AUC $\approx$ 0.524), performance remains slightly above random classification. Such results are typical in behavioral modeling contexts where state transitions are influenced by numerous latent factors beyond observed exposure variables.

The relatively high recall values indicate that when aggression occurs, the model demonstrates some ability to capture transition tendencies.

11.4.5. Psychological Interpretation of Parameters

Estimated parameters provide insight into behavioral mechanisms:

- α (Baseline Activation): Reflects inherent aggression tendency independent of exposure.
- β (Stimulus Sensitivity): Indicates responsiveness to aggressive environmental exposure.
- δ (Recovery/Decay): Captures the rate at which aggression dissipates.

Higher stimulus sensitivity in the male model aligns with lower Granovetter thresholds, suggesting males are more reactive to aggressive contexts. Conversely, the elevated female threshold indicates greater behavioral inertia and resistance to aggression escalation.

11.4.6. Simulation Insights

Simulated behavioral trajectories reproduce intermittent aggression bursts separated by dominant calm periods. Male simulations display slightly higher variability, while female trajectories appear more subdued and stable.

These simulations reinforce the empirical finding that aggression is episodic and rarely persistent across both genders.